Cost Effective Housing Techniques

Unit II: Low Cost Resilient Housing Technology

Syllabus

Development and adoption of low cost resilient housing technology: introduction - adoption of innovative cost effective construction techniques - adoption of precast elements in partial prefabrication - adopting total prefabrication of mass housing in India - general remarks on precast roofing / flooring systems - economical wall systems - single brick thick load bearing wall - 19 cm thick load bearing masonry walls - half brick thick load bearing wall - fly-ash gypsum brick for masonry - stone block masonry - adoption of precast RC plank and joist system for roof / floor in buildings.

1. Introduction

Housing cost can be cut by using **improved construction technology** that saves **materials, time and labour**, and also gives a **resilient** (disaster-resistant, durable) house. Development of such technology in India has been led by **CBRI (Roorkee), SERC (Chennai), NBO, BMTPC, HUDCO Building Centres, IITs** and NGOs.

Principles of cost reduction in technology

- **Save materials**: reduce thickness of walls and slabs, use of **hollow, perforated, cored elements**
- **Replace costly materials** (steel, cement, timber) by local and waste based materials (lime, fly ash, bamboo, stone, soil)
- **Speed**: factory made precast parts reduce construction period and interest on money
- **Standardisation and modular coordination** (100 mm module), repetition of the same unit gives mass production
- **Skill and tools**: simple methods that local masons can learn; **less wastage**
- **Appropriate technology**: suited to local climate, materials, culture, and disasters (earthquake, flood, cyclone)

Resilient housing means

* Strong against **earthquake, wind, flood, fire** - safe life and property
* **Durable** with low maintenance; **thermally comfortable**

* Quick **repair and recovery** after the disaster
* **Sustainable**: low embodied energy and carbon

2. Innovative Cost Effective Construction Techniques

A. Foundation techniques

Technique	Details and saving
Arch foundation	Brick arches between footings carry load of wall; saves 30-40% of masonry and concrete in foundation
Brick (stone) on edge foundation	Bricks laid on edge for light loads; saves 15-20%
Under-reamed pile foundation	Bored pile with 1-2 **bulbs (under-reams)** at base; for **black cotton / expansive soil** and loose soil; pile dia 200-300 mm, length 3.5 m or more; bulb dia 2.5-3 times shaft; **bulbs prevent uplift**
Raft / strip footing	For weak soils and with plinth beam
Rubble trench foundation	Trench filled with stone and lime; good drainage, low cost
Plinth reduced in height of 30-45 cm	Saves filling

B. Walling techniques

- **Rat-trap bond** - bricks laid on edge with a **cavity** in the wall (like rat trap); saves **25% bricks and 10-15% cost**, good thermal insulation, 230 mm wall (same strength); needs mortar 20-25% less
- **Cavity wall, hollow concrete / clay block, solid blocks (stabilised soil blocks, fly ash bricks, AAC blocks)** - big size gives fast work
- **Thin walls: half-brick and 19 cm** walls (see below), and **stone masonry in lime / cement-sand** with bonding courses
- **Ferrocement / precast wall panels, cellular lightweight concrete (CLC)**

C. Roofing and flooring techniques

Technique	Details and saving
Filler slab	Slab with **filler materials (clay pots, tiles, bricks, hollow blocks)** in the tension zone (concrete there is not needed); saves **10–25% cost** of concrete and steel, and improves insulation
Brick arch / Jack arch roof	Shallow brick arches between steel / RCC joists; no shuttering; saves 20–30%
Funicular shell	Shell of funicular (catenary) shape in compression; saves 30–40% concrete and steel
Ferrocement channel / shell units	Thin, light channels spanning 3–4 m, laid side-by-side, finished with screed; saves 30–35% cost vs RCC slab
Precast RC plank and joist	Described later (very popular)
Cored / hollow-core slab	Voids in precast slab reduce weight
Madras terrace, tile roofing on rafters, GI / AC sheet roof	Traditional / lightweight systems
Waffle slab, ribbed slab	Material saved in non-structural zones

D. Other economies

- **Doors and windows: RCC / ferrocement frames and shutters**, steel sections, **frameless doors**, or **plain wooden planks from plantation timber**; use sliding / pivoted shutters; **avoid lintel by arches**; **RCC lintel cum sunshade**
- **Finishing: no plaster** (rat-trap with pointing), **lime-cement mix plaster, cement-wash**; stabilised mud plaster; **neeru finish**; **IPS flooring**, brick floors, **mosaic tiles** only where needed
- **Services: combined plumbing wall**, **PVC pipes**, rain-water harvesting, solar water heaters, **CFL / LED** lighting

3. Adoption of Precast Elements in Partial Prefabrication

Prefabrication is the manufacture of building parts in a **factory or casting yard**, which are then transported and **assembled at site**. In **partial prefabrication** only some elements (lintels, window sills, door frames, planks, joists, stairs, sunshades, beams, columns, chajjas) are precast, and the rest (walls, foundations) are built in the usual way.

Precast element	Notes
Lintels, sunshades, chajjas, sills	Common, cast in mould, cut site work
RC door and window frames	Saves timber; durable; termite proof
Precast RCC planks and joists	For roof / floor (see section 7)
Stair flights, landings	Quick fixing
Precast columns, beams	Frames, moment connections
Ferrocement shells, channels	Roofing
Cover blocks, kerbs, manhole covers, septic tank rings	Small precast products
Hollow concrete and solid blocks, paving blocks	Block making machines

Advantages and limits of partial prefabrication

* **Saves** shuttering, centering (up to 80%), time, and **cement and steel** by quality control
* Less site labour and skill; **better quality** (cured in yard); **fast** construction
* Components can be **stored and reused**; construction continues in rains
* **Limits:** transport and **crane need**, joints (weak points), initial investment for moulds, and demand must be big enough

4. Total Prefabrication of Mass Housing in India

In **total (full) prefabrication** the whole building (walls, floors, roof, stairs) is made from factory-produced **large panels or boxes**, assembled by cranes and joined at site. It is useful for **mass housing** (hundreds of identical units), because the high cost of the plant is spread across many units.

Systems of prefabrication

System	Description
Large panel system	Storey-high **wall panels and floor slabs** cast in a plant, erected by crane; dry or wet joints
Frame (skeleton) system	Precast **columns and beams** with floors and non-load-bearing panels
Box (modular) system	Complete room units (boxes) made, stacked at site
Slab-column (lift slab) system	Slabs cast at ground and lifted
Mixed / hybrid system	Cast-in-situ core with precast floors and walls

Experience in India

- **Early attempts (1960s – 1980s):** HUDCO and NBO, **Hyderabad, Delhi, Mumbai, Pune** large panel projects; **Hindustan Prefab Ltd (HPL, Delhi)**; **Bombay Mhada**; many were stopped due to **high cost, transport problems, and lack of demand continuity**
- **Modern: Tunnel form (Mivan) / aluminium formwork** - monolithic casting of walls and slabs; **precast light weight panels (EPS cement sandwich panels, GFRG)**; **Light Gauge Steel Framed (LGSF) structure**; **Light House Projects (LHP) under GHTC-India 2019-21**: Indore (precast sandwich panel), Rajkot (monolithic concrete with tunnel forming), Chennai (precast concrete system), Ranchi (3D volumetric precast), Agartala (steel frame with GFRG panels), Lucknow (PVC panel + steel frame)
- **Prefab policy:** BMTPC promotes **Performance Appraisal Certification Scheme (PACS)** for new materials and technologies, **Technology Sub Mission under PMAY(U)**

Requirements for success of total prefab

* **Large scale and continuous demand** (at least 500-1000 units / year per factory)
* **Standardisation, modular design** and good planning of **joints** (earthquake and weather proof)
* **Transport and cranes; plant within 30-40 km** of the site; skilled team and quality control
* Government support: **bulk orders**, finance, and approval of codes

Merits and demerits of total prefabrication

Merits	Demerits
Fast (40-50% time saving), **less labour**, **superior quality**	High **initial cost** of plant, moulds and cranes
Less wastage, **less water** at site, clean site	**Transport** cost of heavy elements; roads
Repetition gives economy for large projects	Monotony in design, less flexibility; **difficult joints**
All-weather work; supervision easy	Needs skilled engineers; **uneconomical for small jobs**

5. General Remarks on Precast Roofing and Flooring Systems

- Roof and floor costs about **20-25% of the building cost**, so precast systems are widely used
- **Requirements:** structural strength to carry load and **diaphragm action** (earthquake forces must be transferred to walls), **light weight**, easy handling by **2-4 persons**, no shuttering and propping (or only light props), **thermal insulation, fire resistance**, good appearance of the ceiling, and **proper bearing** on walls (min **90-100 mm**) with a **bond beam**
- **Types:** (1) **RC plank and joist**, (2) **channel units** (cored, ferrocement), (3) **waffle / ribbed panels**, (4) **hollow-core planks and double-T slabs**, (5) **precast RC cored units (Cellular)**, (6) **L-panel, Jack arch with joists, brick panels (reinforced brick panel)**, (7) **Precast hollow clay tile slabs** (Madras terrace type)
- **Joining:** use **cast-in-situ screed (35-50 mm) over the units with mesh**, grouting of joints, **shear keys**, and tying with the **RCC bond beam** at roof level, for earthquake safety

6. Economical Wall Systems

Walls take 25-35% of the cost of a house. So economy of walls is the first target. Wall thickness is reduced where only **light loads** (single / double storey) are carried.

A. Single brick thick load bearing wall (230 mm)

- The **standard wall** of 230 mm (9 inch) in **English bond** in cement mortar **1:6**; used for **load bearing** up to **3-4 storeys**
- Economy by using **rat-trap bond** (saves 25% bricks), mortar 1:6 or 1:8, **no plaster**, or **lime-cement** mortar
- Strength: brick of 75 kg/cm^2 (7.5 N/mm^2) class gives permissible stress about **0.5 N/mm^2 (IS 1905)**
- **Slenderness ratio** limits: effective height / thickness not more than **27** (for 230 mm wall = about 3.1 m height)

B. 19 cm thick load bearing masonry wall

- A wall of **19 cm** thickness is made from the **brick of nominal size 19 x 9 x 9 cm** (modular) laid with **brick on header** (one brick = 19 cm), with **1:6** mortar
- Used for **single-storey and two-storey** buildings; **saves 17% of brickwork** vs 230 mm (so 15-20% on wall cost), and reduces dead load, increases floor area
- Strength is enough for **low rise (up to 2 storeys)**: slenderness ratio checks; must be used with **RCC bands** at plinth, lintel and roof level, and **vertical reinforcement at corners** in earthquake zones
- Developed by **CBRI Roorkee**; for Zone III and below; **mortar 1:6** to 1:4; height of wall not more than **3.0 m**

C. Half brick thick load bearing wall (115 mm)

- Wall of **half brick** (stretcher bond) 11.5 cm thick: used as **partition walls** and also load bearing in **single storey buildings** in light roofing, with **short span (3.0 - 3.5 m)**
- Needs **stiffening** by **buttresses / piers** at 1.2 m, **cross walls**, vertical bars, or **RCC / brick-on-edge bands**; panel size limited; **height up to 2.5-3 m; length limit 3.5 m between cross walls**; mortar 1:3 to 1:4
- Saves **50% of bricks and mortar** as compared to 230 mm wall; reduces **dead load (good for seismic)**; but poor in thermal insulation and sound, and **weak in lateral load**, so used with **reinforced brickwork** (bars in every 4th course, **Brick-bat bands**) in seismic area
- Where loading is lighter, **cavity walls** may use 115 + 115 mm leaves with 50 mm gap

Wall type	Thickness	Use	Saving
Standard brick wall	230 mm	Multi-storey load bearing	—
Rat-trap bond	230 mm (cavity 75 mm)	Up to G+2	25% brick, 10-15% cost
19 cm wall	190 mm	Up to 2 storeys	15-20% of walling cost
Half brick wall	115 mm	Partition, single storey	Up to 50% bricks
Fly ash / AAC blocks	100-200 mm	Infill / load bearing	Light, thermal comfort

D. Fly ash - gypsum (FAL-G) bricks for masonry

Fly ash (a waste from thermal power stations) is mixed with **lime and gypsum**, producing **FaL-G bricks**. A pozzolanic reaction forms **calcium silicate and sulphoaluminate** hydrates that give strength.

FaL-G brick facts

* Mix: **fly ash 50-60%, lime 10-20% (or cement), gypsum 5-10%, stone dust / sand 20-30%**; moulded under **pressure**, **water cured** (not burnt, saves coal and top soil)
* **Size 230 x 110 x 75 mm (or modular)**, **compressive strength 7.5 - 12 MPa (IS 12894)**, water absorption below 20%
* **Light weight** (1.4-1.6 g/cc), smooth, accurate; **plaster can be thin or omitted**; **10-20% cheaper** than burnt clay bricks
* **Environment:** uses an industrial waste; **no top soil** is lost; less CO_2; **Fly ash brick use mandatory** (MoEF notification) within 300 km of thermal plants

- **Fly ash blocks (hollow / solid) and fly ash gypsum thick walls** are used for load bearing and infill walls: the **thick (230 mm) wall** from large FaL-G bricks provides **good thermal insulation**
- **Precautions:** adequate curing (7-14 days), avoid dry shrinkage, use cement-lime mortar, **do not use in contact with water** without damp-proof course

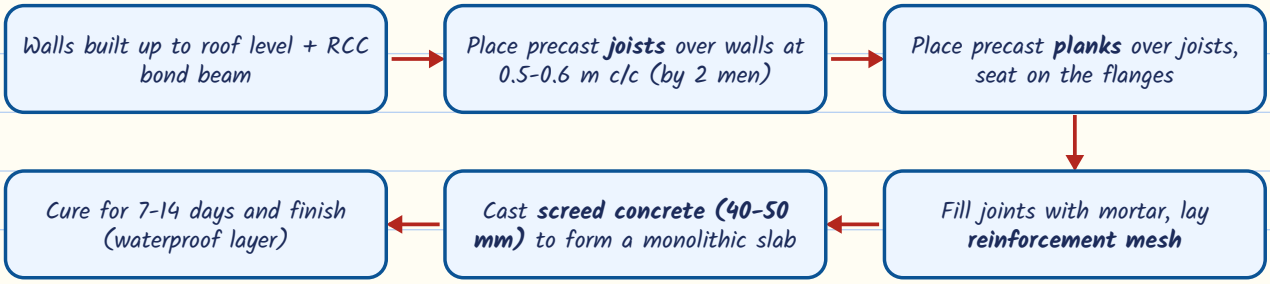
E. Stone block masonry

- Where **stone** is available (hilly and plateau regions), **dressed stone blocks** (large regular blocks, say **300-450 mm long, 200-250 mm high, 200-300 mm thick**) are used instead of random rubble; **quick laying, less mortar (saves 30-40% mortar)** and fewer joints; **walls may be 300-400 mm** thick; **stone is available free** at local quarries

- **Types:** ashlar masonry, **squared rubble**, **block-in-course**; **cut stone blocks of 3 cm face**; **laid with cement-sand 1:6** or lime mortar, with **through stones** at every 1.2 m, **corner stones** and **horizontal bands**
- **Stone block** can be **hollow stone (cut out voids)**, or **concrete blocks made of stone chips**; with **bonding and RCC bands**, they work well in earthquake and give a **cool house**
- **Economy:** local stone, less mortar, strong, **durable**; **disadvantage:** heavy, high transport, low thermal insulation if thin; needs skilled stone cutters

7. Precast RC Plank and Joist System for Roof / Floor

A **precast RC plank and joist** system was developed by **SERC (Chennai) and CBRI** for **low-cost roofing** of buildings with spans up to **3–4 m**. The roof has **precast joists (small beams)** laid at about 500–600 mm centres, with **precast planks** (thin slabs) resting on joists; then a **cast-in-situ screed (30–50 mm)** with nominal reinforcement.



Item	Typical details
Planks	Size **300 x 500 x 40–50 mm**; M20 concrete, **2 bars of 4–6 mm**; weight 10–30 kg: carried by one man
Joists	Size **100–130 x 150–200 mm**; **T or L section** with stirrups, reinforcement 2 bars of 8–10 mm; length up to 3.5 m
Screed	**1:2:4 concrete or M20**, 40–50 mm thick with **weld mesh / 6 mm bars @ 200 mm**
Span	Up to **3.6 m**
Saving	**25–30% on cost** and 30–35% in steel and cement vs. RCC slab; no shuttering, only light props for joists (removed after 14 days)
Variants	Joist + **hollow clay tile / pot / filler block** system; **planks of ferrocement, brick, or lightweight**; joist + stone slab (precast)

Merits and precautions

* **Merits:** saves shuttering and centering, quick, low cost, works by local labour, **good quality** of precast elements, easy repair
* **Precautions:** proper **bearing (at least 100 mm)**, joists should be **propped** until the screed gains strength, **joints** filled well, **tie the roof with bond beam** for earthquake, **waterproofing** treatment, edge reinforcement

8. Cost Saving: A Worked Comparison

Example: comparison of RCC slab vs. plank and joist roof for 3 m x 4 m room

RCC slab (100 mm): concrete = 12 x 0.1 = 1.2 m^3; steel about 60 kg (50 kg/m^3 x 1.2) + shuttering (12 sq.m).

Plank & joist: planks 0.8 m^3 + screed 0.5 m^3 + joists - cuts concrete by 30% and steel by 40%; no shuttering (props only) => cost saving of about **25-30%** and time saving of 2-3 weeks.

9. Modular Coordination and Standardisation

- **Modular coordination** = dimensioning of building and its components in multiples of a basic **module M = 100 mm** (IS 1233), so that parts from different makers fit together without cutting
- Preferred multi-modules: **3M (300 mm), 6M, 12M**; **brick size 190 x 90 x 90 mm** (modular) vs 230 x 110 x 75 mm (traditional)
- **Benefits:** less wastage, reduces **site cutting**, free interchange, **mass production** of doors, windows, planks, **easier design and estimating**
- **Standard components:** door 1000 x 2100 mm, window 1000 x 1200 mm, **ventilators 600 x 450 mm**, stair width 0.9-1.0 m, rise 150-170 mm, tread 250-275 mm

10. Resilience: Safety Features in Low Cost Technology

Feature	Why / how
RCC / reinforced brick bands	Plinth, lintel, roof (and sill) bands tie walls like a **box** against earthquake (IS 4326)
Vertical steel at corners and jambs	Prevents separation of walls at corners
Regular, symmetrical plan, light roof	Less seismic force, no torsion
Wall length / opening limits	Openings not more than 1/3 of wall length, 600 mm away from corners
Cyclone / wind	Hip roof with slope 30 degrees, **tie-down straps**, strong foundations, low height
Flood	Raised **plinth**, stilts, water resistant materials, stabilised blocks
Fire	Non-combustible materials, fire-retardant treatment of thatch
Quality precast joints	Wet joints, shear keys, **dowels and bolts** transfer the diaphragm forces

- **Precast units must be tested** for load (transverse strength) and dimensions; **cube strength** of concrete (M20 or more), **curing 14 days** (water curing), **stacking on level ground**, lifting at the marked points
- **Cost control** by **bulk purchase, sharing mould sets (use 50-100 times), semi-skilled labour, and local production through Building Centres**

11. Quick Revision Points

Remember for exam

* Rat-trap bond saves about 25% bricks; filler slab saves 10-25%; plank and joist saves 25-30% (no shuttering)
* Under-reamed piles are used in expansive soils; arch foundation saves 30-40%
* 19 cm wall saves about 17% of brickwork; half-brick wall needs stiffening and bands
* FaL-G brick = fly ash + lime + gypsum (+ sand), cured not burnt, 7.5-12 MPa
* Total prefabrication needs mass demand: large panel, frame, box systems; Light House Projects (2021)
* Modular coordination (100 mm module), standardisation and bond beams are the essentials of safe precast

Important questions

1. Explain the innovative cost-effective construction techniques for foundation, walls and roofs.
2. Describe partial prefabrication with precast elements and its advantages.
3. Discuss the total prefabrication of mass housing in India with the types of systems.
4. Write short notes on precast roofing and flooring systems.
5. Explain economical wall systems: 230 mm wall, 19 cm wall and half-brick wall.
6. What are fly-ash gypsum bricks? Explain their composition and uses.
7. Explain stone block masonry and its advantages for cost-effective housing.
8. Describe the construction of the precast RC plank and joist roof / floor system with a neat sketch.